BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 233 · Waterfall, Agile and Extreme Programming

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which OCR pseudocode function reads the next line of a text file?

Q2. What is a function parameter?

Q3. Why use boundary-value testing?

Q4. If requirements frequently change, what is the risk of freezing all requirements before implementation?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Waterfall — Primarily sequential software-development life cycle.

Agile — Iterative approach using feedback and incremental releases.

XP — Extreme Programming, an agile methodology with disciplined engineering practices.

sprint — Short time-boxed development iteration.

test-driven development — Write an automated test before writing the production behaviour.

New learning

Waterfall is sequential — Agree requirements, then design, implementation, testing and deployment. Revisiting an earlier stage can be costly.

Agile delivers incrementally — Each short iteration produces working software and seeks feedback, so priorities can change between iterations.

Extreme Programming (XP) — XP is an agile methodology emphasising practices such as test-driven development, pair programming, refactoring and continuous integration.

Worked example

A hospital requests software with a carefully agreed specification and formal approvals. Why might Waterfall suit it?

1. Predict first — Requirements and approval milestones can be documented before later stages proceed.

2. Model and explain — Waterfall offers clear documentation and control, but late changes can require expensive rework.

We Do · Practise with guidance

A pupil revision app changes after each group of students tries it. Which approach fits and why?

```
SPRINT 1: quiz → feedback
SPRINT 2: revision history → feedback
SPRINT 3: accessibility improvements
```

Predict first

Changing needs and quick feedback favour Agile iterations over a rigid up-front plan.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Model and explain

A usable quiz is released first, then revision history and accessibility features improve in later sprints.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State a defining characteristic of Waterfall, Agile and XP.

Question 2

A mobile app company receives new customer requirements each fortnight. Recommend a methodology and explain two advantages and one disadvantage.

Question 3

Explain how test-driven development and continuous integration in XP could reduce defects, and why neither guarantees fault-free software.

Exit Ticket · Answer before checking

Agile means software is never planned or tested.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. readLine() on an opened file object.

Q2. A named input received by a subroutine.

Q3. Defects often occur at the limits of valid ranges.

Q4. Large amounts of design/code may need reworking when changes emerge.

You Do pages · Independent questions

Q1. [3] Waterfall follows largely sequential planned stages [1]; Agile develops in short iterations with feedback [1]; XP uses practices such as pair programming and test-driven development [1].

Q2. [5] Recommend Agile [1]; frequent releases/feedback [1]; priorities can change between sprints [1]; customer involvement helps reveal errors early [1]; less predictable full scope/timescale [1].

Q3. [6] Write a failing test before implementation [1]; implement just enough to pass [1]; refactor with tests [1]; integrate changes frequently [1] to detect conflicts early [1]; tests cannot cover every case and may themselves be incomplete [1].

Exit Ticket · Model answer

False. Agile uses planning and testing repeatedly across short iterations.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding